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Potential Modifications or Additions to the ASME Section VIII Rules to Provide Greater Confidence in the Design of High Temperature Nuclear Components

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Potential Modifications or Additions to the ASME Section VIII Rules to Provide Greater Confidence in the Design of High Temperature Nuclear Components

Applied Materials Division

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EXECUTIVE SUMMARY

Section VIII of the American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel Code (BPVC) provides rules for construction of pressure vessels. Advanced reactor vendors have expressed interest in using the Section VIII rules to design their components, but the rules have not been endorsed by the Nuclear Regulatory Commission (NRC). This report develops and documents supplemental design criteria to the existing ASME BPVC Section VIII rules to provide increased confidence in the long-term reliability of high temperature components. The methods address potential limitations in the high-temperature Section VIII rules related to definition of the allowable stresses, the lack of high temperature fatigue data, the lack of creep-fatigue design rules, and the strength of weldments in the creep regime. As much as possible, the proposed supplemental design checks are material agnostic, though the analysis in this report relies on the high temperature test data collected for the ASME Section III, Division 5, Class A materials. Where feasible, the report identifies criteria that could be generalized to broader material categories or summarizes the test data that would be needed to extend the methods to other materials.

The report provides a quantitative estimate of the difference in component reliability inferred from the different definitions of the allowable stresses applied in Section VIII and Section III, Division 5. While this comparison is incomplete, as the allowable stress criteria are only one aspect of the complete design method for both sections of the Code, it does provide some estimate of the difference in expected reliability between components designed according to the respective rule sets.

The report then provides criteria for demonstrating special design and operating conditions where the differences between the Section VIII and Section III, Division 5 design rules are less significant or insignificant, meaning a component could be reasonably designed to either set of rules without needing to evaluate in detail the differences between the design methods. The specialized criteria considered in the report are:

1. Negligible load – the structural demand on the component is so small relative to the material strength that the component will pass any of the design rules considered here.
2. Negligible creep – while the component operates in the elevated-temperature regime, creep effects are negligible.
3. Negligible fatigue – while the component may operate under cyclic load, fatigue effects are negligible.

Finally, the report considers critical differences between Section III, Division 5 and Section VIII related to weld design and inspection.

The report also notes the ongoing effort by the cognizant ASME Code Committees to add detailed creep-fatigue design methods to ASME Section VIII, which may eventually obviate the simpler, supplemental approaches described here.

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1 Introduction

Reactor vendors have expressed interest in utilizing the Code rules in Section VIII of the American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel Code (BPVC) for component design. Unlike Section III, Division 5 of the ASME BPVC, the Section VIII rules have not been reviewed or endorsed by the Nuclear Regulatory Commission. Previous work identified several key differences between the Section VIII rules for high temperature design and corresponding rules in Section III, Division 5, which covers the design and construction of high temperature reactor components [1]. These key differences include:

1. Section VIII allowable stresses generally are higher than the Section III, Division 5 allowable stresses.
2. The Section VIII fatigue design methods do not provide data for high temperature cyclic design, except via an exemption for components where comparable successful service experience can be identified. Section III, Division 5 provides high temperature fatigue design rules and data.
3. Section VIII does not directly account for the reduced strength of weldments in elevated-temperature conditions. Section III, Division 5 accounts for this difference in strength with weld stress rupture factors.
4. The Section VIII rules for cyclic design do not consider creep-fatigue interaction. Section III, Division 5 provides extensive creep-fatigue design rules.

This report describes several additional design checks and methodologies which could supplement the ASME Boiler & Pressure Vessel Code Section VIII rules to provide greater assurance of long-term structural reliability for nuclear components operating in the elevated-temperature regime.

The focus of this report is on standard operating conditions, exemplified in the ASME Section VIII rules by the Design Condition and, in Section III, Division 5, the Service Level A conditions. In general, in Section VIII, Division 1 and for the allowable stress design in Section VIII, Division 2, the rules do not consider transient conditions. The focus of Section VIII is on standard, steady state conditions, with some exceptions in Division 2 for detailed fatigue design by analysis. Off-normal conditions are, by definition, lower probability events compared to the standard operating conditions. Except for seismic considerations, Section VIII does not consider off-normal events during the component design, whereas Section III, Division 5 considers the off-normal events by providing higher allowable stresses. This is done through the use of service levels within the Section III, Division 5 methodology. The focus of this report is therefore on standard conditions, both to facilitate the comparison between the Section III and Section VIII rules and to avoid specific judgement on what an appropriate design margin might be for less frequent events.

Chapter 2 documents the difference in component reliability inferred from the different definitions of the allowable stresses. This analysis provides the probability of failure via creep or plastic rupture for a given design life, under steady load, implied by the contributing material properties and factors making up each allowable stress definition. This chapter provides insights into quantifying the design margin for a given component.

Chapter 3 provides a negligible load method for high temperature component design. The idea is that if the loads on the component, including pressure, thermal, and mechanical forces, are low enough that component would pass the Section III, Division 5 and Section VIII design criteria. Such a component then could be safely designed against either set of rules. The main result of the analysis in this chapter is a simple criterion defining a region of negligible load for many materials but excluding creep-strength enhanced ferritic steels like Grade 91.

Chapters 4 and 5 address the lack of creep-fatigue design criteria in Section VIII by providing criteria for identifying components operating in negligible creep and negligible fatigue conditions. Such a component will not be subject to creep-fatigue interaction because one of the two contributing damage mechanisms, i.e. creep or fatigue, is negligible. Again, this chapter provides generic criteria applying to components constructed from a variety of materials. Chapter 4 also addresses the lack of high temperature fatigue data in Section VIII by providing recommendations for where the required fatigue strength data could be found.

Finally, Chapter 6 addresses the strength of weldments operating in the creep regime. Section VIII currently does not reduce the strength of weldments in high temperature components, despite extensive past research demonstrating that even weldments that are overmatched at room temperature for time-independent properties will be undermatched in creep conditions for time-dependent properties [2-18]. This difference in strength is addressed in the Section III, Division 5 (and Section I) rules by factors applied to the base material allowable stress and creep strength in the weldment regions. Chapter 6 briefly reviews the literature on the creep-strength of weldments and provides recommendations for how to implement weld strength reduction factors in the Section VIII design and where suitable factors covering a variety of Section VIII materials might be found.

2 Differences in Margin Between the Allowable Stresses

Section VIII, Division 1 and Division 2 and Section III, Division 5 use different criteria in constructing the allowable stresses that underpin the design method for evaluating the component against the load-controlled stress limits. These limits are designed to guard against plastic collapse and long-term creep failure under steady loading conditions.

Section VIII, Division 1 and Section VIII, Division 2 each rely on a single allowable stress set with an indefinite life philosophy. This means that both Section VIII Divisions do not limit the design life of the component, instead expecting, in principle, it could continue to operate indefinitely. The values of the allowable stresses for time-dependent creep properties are however based on extrapolated 100,000-hour data. Section III, Division 5 has two separate allowable stresses, S_o which is defined nearly identically to Section VIII, Division 1, and S_{mt} , defined with a design life philosophy. A design life philosophy means the component is assigned a definite target service life (the design life) and the time-dependent creep properties in the allowable stress are based off this duration. At the end of the design life the component would either be removed from service or a life extension evaluation conducted to justify additional service time.

Table 2.1. Allowable stress criteria.

	III-5 S_o	III-5 S_{mt}	IID, Table 1 S	IID, Table 5 S
Approach	Indefinite, 100,000 hour	Design life	Indefinite, 100,000 hour	Indefinite, 100,000 hour
Tensile	$0.67 \sigma_{y,min}$	$0.67 \sigma_{y,min}$	$0.67 \sigma_{y,min}$	$0.67 \sigma_{y,min}$
	0.67 to $0.9 S_y$	0.67 to $0.9 S_y$	0.67 to $0.9 S_y$	0.67 to $0.9 S_y$
	$0.285 \sigma_{u,min}$	$0.333 \sigma_{u,min}$	$0.285 \sigma_{u,min}$	$0.417 \sigma_{u,min}$
	$0.285 S_u$	$0.333 S_u$	$0.285 S_u$	$0.417 S_u$
Creep	$1.0 \sigma_{minrate}$	$1.0 \sigma_{1\%}$	$1.0 \sigma_{minrate}$	$1.0 \sigma_{minrate}$
	$0.67^1 \sigma_r$	$0.8 S_{tc}$	$0.67^1 \sigma_r$	$0.67^1 \sigma_r$
	$0.80 S_r$	$0.67 S_r$	$0.80 S_r$	$0.80 S_r$
Other	Currently limited to the lowest value of S_{mt} for the given temperature			

¹ Strictly F_{avg} with $F_{avg} = 0.67$ below 815° C and determined from the slope of the log time-to-rupture versus stress plot at 100,000 hours so that $\log F_{avg} = 1/n$, but not greater than 0.67, with n the slope.

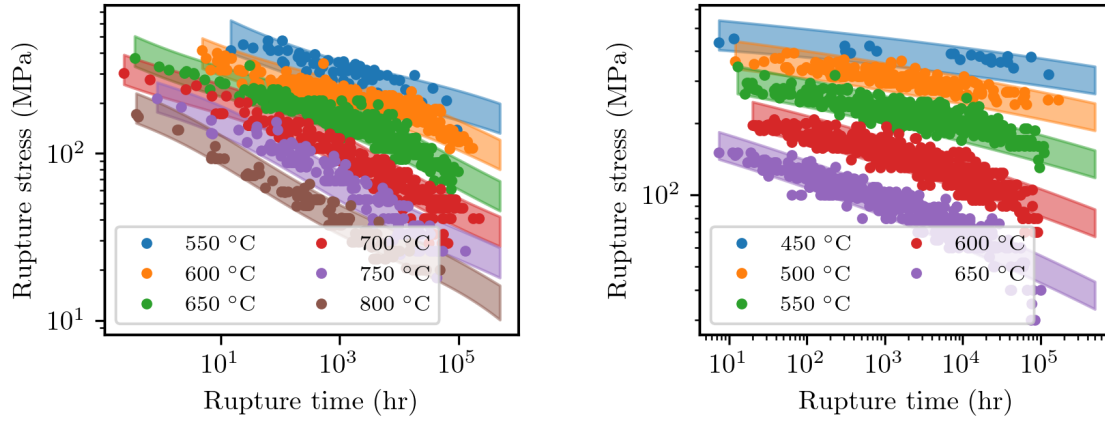
Table 2.1, reproduced from [1], summarizes the criteria underlying each allowable stress. In this table $\sigma_{y,min}$ is the specified minimum (room temperature) yield strength, $\sigma_{u,min}$ is the specified minimum (room temperature) ultimate tensile strength, S_y is the ASME Section II, Part D yield strength, S_u is the ASME Section II, Part D tensile strength, σ_r is the average strength to cause creep rupture in the given time, S_r is the minimum strength to cause creep rupture in the given time, $\sigma_{minrate}$ is the average stress causing a creep rate of 1%/100,000 hours, $\sigma_{1\%}$ is the average stress to cause a total strain of 1% in the given time, and S_{tc} is the minimum strength to cause the onset of tertiary creep in the given time. Note that the allowable stress for Section VIII, Division 1 is provided in Section II, Part D, Table 1, and for Section VIII, Division 2 in Section II, Part D, Table 5.

The different philosophies related to time-dependence and the different definitions of the allowable stress suggest a natural question: what is the design margin of each of the allowable stress definitions? The margin can be defined in terms of material reliability: the marginal probability distribution of material failure over time at the allowable stress for a given temperature.

The first step toward determining margin is to define failure. The allowable stresses all include metrics not directly related to material failure, such as the creep rate/accumulated strain and the yield stress. For the purposes of this chapter, the authors define failure as the material exceeding its actual creep rupture strength. This criterion encompasses the material tensile strength – the creep strength must be less than the tensile strength – but excludes non-failure criteria like creep deformation or material yielding.

Next, it is recognized that the rupture strength of the material is stochastic and that a statistical model for the creep rupture strength is needed. Many approaches are possible, but the authors elected to use the technique described in [2, 3]. This method applies Gaussian Process Regression (GPR) to creep rupture data in a way that retains the basic time-temperature dependence of a Larson-Miller model. This framework also accounts for heat-to-heat variation in the Larson-Miller batch parameter, C , by treating it as a random variable with a normal distribution.

Figure 2.1 plots statistical Larson-Miller diagrams for statistical rupture models of this type for Grade 91 and 316H. The models adequately capture both the average and scatter in the creep rupture data. The Grade 91 dataset includes 2,046 entries, the 316H dataset includes 1,764 entries.



(a) 316H

(b) Grade 91

Figure 2.1. Comparison between the statistical model for creep rupture and test data. The points are experimental data, and the shaded regions are 90% prediction intervals for the model.

The GPR statistical model provides the probability of the log of the rupture stress as a function of the log of the rupture time and the absolute temperature in terms of a multivariate normal distribution defined by:

$$p(\log \sigma \mid \log t_r, T) = N\left(\mu_{GPR}, \frac{\sqrt{\text{diag } \mathbf{K}_{GPR}}}{\sigma \ln 10}\right)$$

where σ is the uniaxial rupture stress, t_r is the rupture time, T is the absolute temperature, μ_{GPR} is the mean rupture stress and $\mathbf{K}_{GPR}$ is the covariance of the Gaussian process at that combination of stress, temperature, and time. The reliability of a particular allowable stress for a given temperature and time is defined as

$$R = 1 - p(\sigma < \sigma_{allow} \mid t, T)$$

with σ_{allow} the allowable stress. For visualization, the probability of premature failure is plotted and defined as

$$P = 1 - R = p(\sigma < \sigma_{allow} \mid t, T)$$

literally the probability that the material fails by creep rupture for stresses less than the allowable stress.

The following four cases are considered:

1. S for Section VIII, Division 1 (nearly the same as S_o for Section III, Division 5).
2. S for Section VIII, Division 2.
3. S_{mt} for Section III, Division 5.
4. The lesser of S_o and S_{mt} for Section III, Division 5.

The value of case 1 gives the reliability of the allowable stress for Section VIII, Division 1, case 2 gives the reliability for Section VIII, Division 2, and case 4 gives the reliability for the entire Section III, Division 5 method. The authors omit case 2 for Grade 91 because Table 5 in Section II, Part D does not yet contain allowable stresses for Type 2 material. Case 3 is included for comparison purposes only, and in case future versions of Section III, Division 5 remove the additional S_o requirements.

Figure 2.2 plots the probability of premature failure implied by the four different allowable stresses for 316H as a function of time, considering six different temperatures. Figure 2.3 is a similar plot for Grade 91.

Neither Section VIII nor Section III, Division 5 give target reliability values for metallic materials. The margin in the Code is largely based on experience and engineering judgement. For graphite materials, Section III, Division 5, Subsection HHA provides allowable failure probabilities between 10^{-4} and 10^{-2} depending on the type of loading and the Structural Reliability Class of the component, which relates to its safety significance. The lower allowable failure probabilities are used for design loadings and Service Level A type conditions.

It is not within the scope of this report to determine an appropriate level of reliability for high temperature components. However, the failure probabilities in Figure 2.2 and Figure 2.3 span many orders of magnitude, making some general comparisons between the two materials, different temperatures, and different design lives possible without delving into appropriate design margin and component reliabilities. As such, the following compares the materials and the performance of the materials at different conditions to each other and makes some general assessment of the order of magnitude of the assessed probabilities of premature failure as either conservative (less than 10^{-4}) or potentially unconservative (greater than 10^{-2}). These categories are based on analogy to the values used for graphite components under design conditions.

Looking first at the higher temperatures, starting at subfigure (b) for each series of figures, the following common trends are observed:

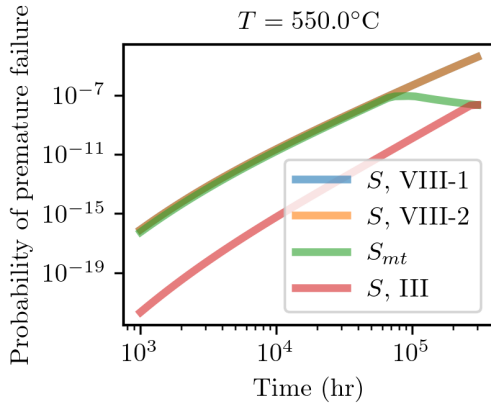
1. The probability of premature failure for the Section VIII allowable stresses increases monotonically with time (i.e. the reliability decreases monotonically). This reflects the indefinite life concept: using a fixed time to determine the allowable stresses means the reliability will decrease with time, as the actual rupture strength of the material decreases.
2. The reliability for S_{mt} is approximately constant as a function of time. Again, this reflects the design life concept inherent in this allowable stress: the values adapt to the time under consideration. The values are only *approximately* constant because:
 - a. other criteria, like time to 1% strain, may control in certain time/temperature regions;
 - b. there are discrepancies between the rupture database used here and the data used in setting the Code allowable stresses;
 - c. the time-independent criteria (yield and tensile strength) control at cooler temperatures, as discussed below.
3. The composite allowable stress for Section III, Division 5 (the lesser of S_o and S_{mt}) switches behavior at a critical time. For shorter times, the overall reliability is monotonically increasing, following the indefinite life S_o values, but then switches to approximately constant, reflecting the time-dependent values of S_{mt} .

4. In the region where S_o controls over S_{mt} the reliability of Section III, Division 5 approximately follows that of Section VIII, Division 1. This reflects the fact that the definition of S_o approximately mirrors that of S for Section VIII, Division 1. The discrepancies (which are small in terms of the allowable stress, but can be large in terms of the probability values) are caused by differences in the material database used to calculate the allowable stresses and the extra criteria added to the definition of S_o in Section III, Division 5, notably that the values are cutoff by the value of S_{mt} for times longer than 100,000 hours if they fall below the calculated value of S_o .
5. For 316H, the reliability inherent in the values of S for Section VIII, Division 1 and Division 2 is compared. They are nearly the same for these temperatures, as the difference in definition is only for the time-independent yield and tensile strength criteria.
6. Though the trend is not strictly monotonic, in general the Section VIII allowable stresses at the longest times decrease in reliability as temperature increases. Presumably this reflects a sharper drop off in rupture stress beyond 100,000 hours at higher temperatures compared to lower temperatures.
7. There is a strong material dependence in the values of reliability for the Section VIII methods at longer design lives (and for all four sets of allowable stresses in general). The Grade 91 allowable stresses have a lower probability of premature rupture than the 316H allowable stresses.
8. For Grade 91, at realistic use temperatures between 550 and 600°C and longer design lives, the probability of premature failure for the Section VIII methods ranges from approximately 10^{-5} to 10^{-7} , which is the highest among plotted operating conditions.
9. For 316H, at realistic use temperatures between 600 and 700°C and longer design lives, the probability of premature failure for the Section VIII methods approach 10^{-2} , which is potentially unconservative using the limits identified above.
10. As noted above in this list, the reliabilities for S_{mt} are not exactly constant with time and vary somewhat with material and temperature. However, in general, the probability of premature failure remains less than 10^{-7} in all cases, approaching values less than 10^{-10} for most conditions. This is considered to be conservative using the limits identified above.
11. As the composite reliability of the Section III, Division 5 method is bounded by the reliability for S_{mt} , i.e. the reliability of the method is even greater for cases controlled by S_o , the overall reliability of the Section III, Division 5 design method for load-controlled stresses is conservative using the limits identified above.

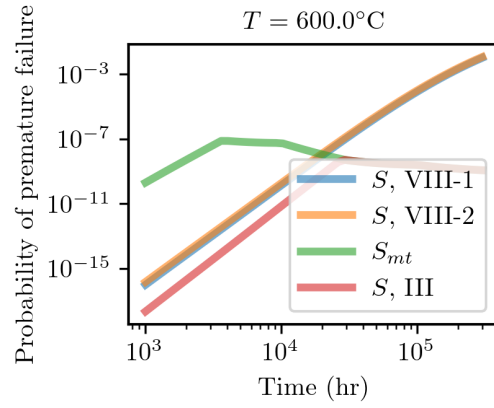
The analysis in this chapter attempts to rigorously quantify the reliability of Section VIII and Section III, Division 5 methods. There are several limitations to the approach:

1. It only considers the allowable stresses underpinning the load-controlled stress limits. It does not consider the other design criteria, including strain accumulation, ratcheting, fatigue, and creep-fatigue. These criteria, particularly creep-fatigue, are often more significant in the design of high temperature reactors than the plastic collapse and long-term creep rupture failure modes encompassed by the allowable stress criteria.

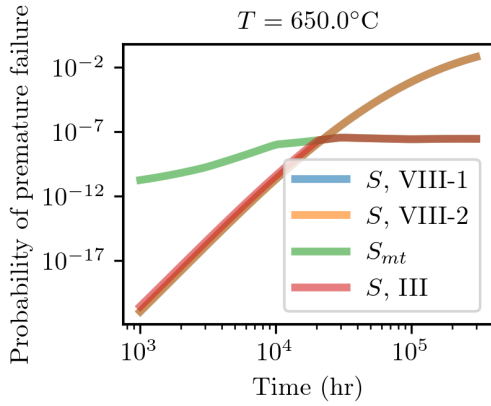
2. It only considers the allowable stress in the context of the test data, i.e. for uniaxial load under controlled, isothermal conditions. This is not fully representative of the multiaxial loads experience by a component in service and does not precisely correlate to premature component failure. A full assessment of the methods, even for only the load-controlled stress limits, would need to consider more representative conditions.
3. The definition of failure in this analysis only considers material failure and not any other design limits like the onset of tertiary creep, 1% strain accumulation, and material yielding.
4. It only considers base metal. Section III, Division 5 provides stress rupture factors to account for the reduced high temperature creep-strength of weldments. These factors should keep the failure probabilities of weldments consistent with the base metal analysis conducted here. Section VIII does not reduce the strength of weldments at high temperatures, which may indicate higher failure probabilities for weldments versus base material when applying those methods.



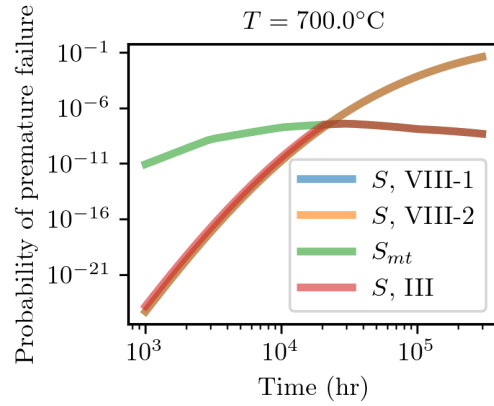
(a) 550°C



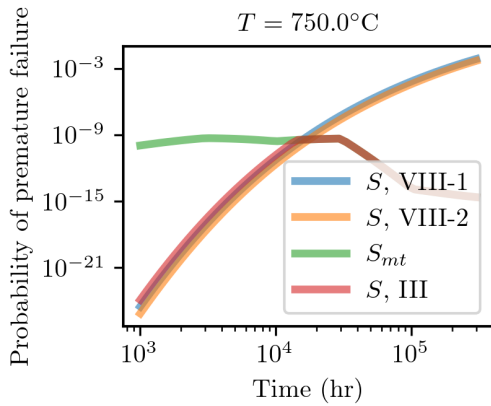
(b) 600°C



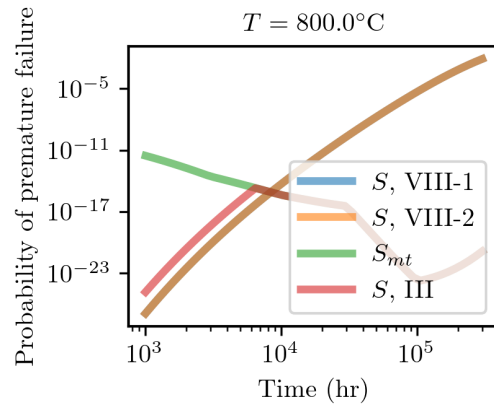
(c) 650°C



(d) 700°C

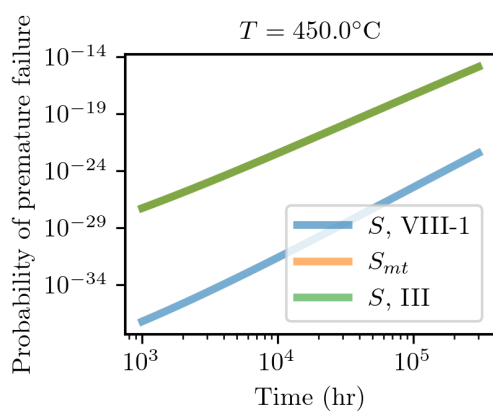


(e) 750°C

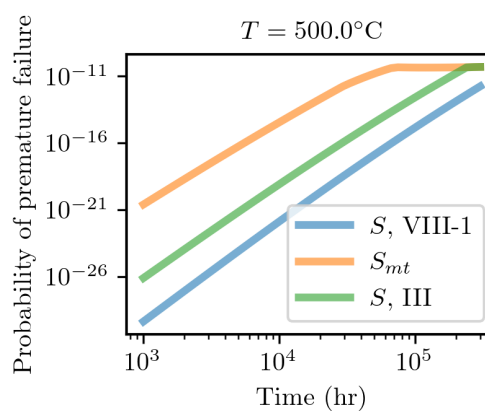


(f) 800°C

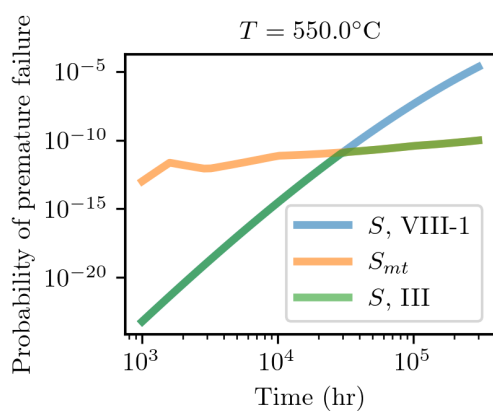
Figure 2.2. Reliability comparison for the allowable stresses for 316H.



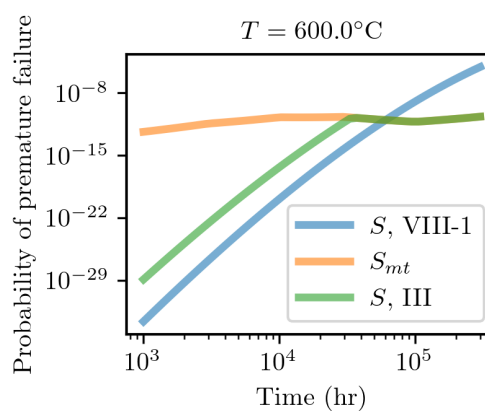
(a) 450°C



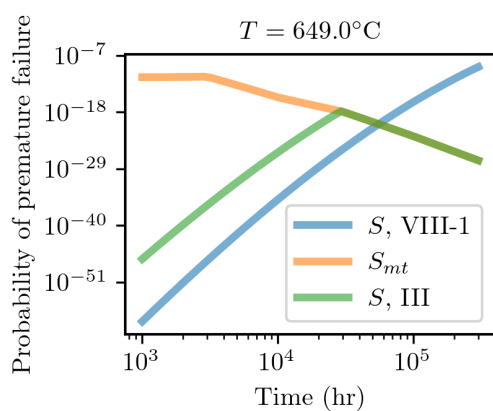
(b) 500°C



(c) 550°C



(d) 600°C



(e) 649°C

Figure 2.3. Reliability comparison for the allowable stresses for Grade 91.

3 Negligible Load

3.1 Background

A previous report [1] quantitatively compared the design envelope of the Section III, Division 5, Class A design rules to the envelope for the following design rules: Section VIII, Division 1; Section VIII, Division 2, Part 4; and Section VIII, Division 2, Part 5. That report analyzes the different design methods by considering a parametric design problem, specifically the Bree cylinder, where the problem geometry, generalized loading (both primary and secondary), cycle hold time, number of transient load cycles, and component design life can all be varied. For each parameterized design and for each set of design rules, the analysis applied the design rules to the problem to determine if that component would be acceptable or not. By repeating this analysis for a wide variety of conditions, the report generated Bree diagram summary figures of the type shown in Figure 3.1 for 316H. These plots show the design envelope for the four sets of design rules for different temperatures, cycle durations, and design lives.

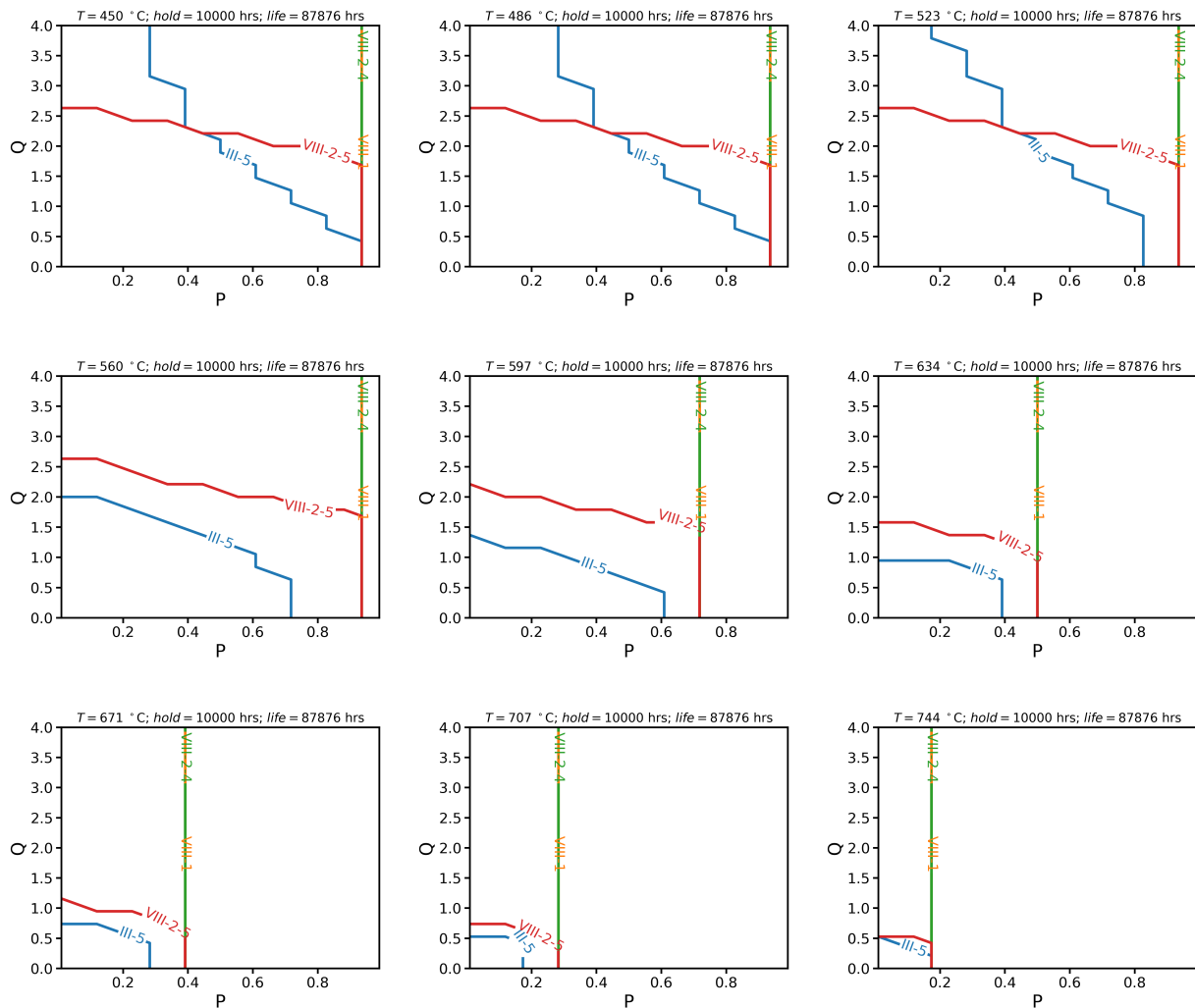


Figure 3.1. Temperature series of design space charts for 316H.

This process provides a design envelope for each set of design rules plotted here in a generalized load space with non-dimensionalized primary load, P , on the x-axis and non-dimensionalized secondary load, Q , on the y-axis. The non-dimensionalized primary load is the membrane pressure stress in the Bree cylinder divided by the yield strength. The non-dimensionalized secondary stress is the maximum thermal stress on the cylinder, again divided by the yield strength. These diagrams normalize the stresses to the ASME design material yield strength S_y at fixed temperature, ignoring the variation of temperature during the load cycle. Designs that fall inside this envelope (i.e. closer to the origin than the line defining the envelope) pass the design rules; those outside fail.

By comparing these envelopes for all four sets of rules a “safe region” can be identified. Designs in this region would pass for all four sets of design rules. In [1], this region is identified as a design space where it would be reasonable to apply the Section VIII rules to a high-temperature reactor component.

The bounds of this region depend on the material type and *all* the parameters defining the design. Generally, the bounds contract as the temperature increases and generally the Section III, Division 5 rules are the most limiting set, defining the envelope. However, there is significant variability in how the envelop behaves as a function of temperature and design life across the designs and materials considered in the analysis.

3.2 Negligible load identification

3.2.1 Base analysis

The results in this section seek to determine a negligible load envelope, strictly defined as the space in which a design passes all four sets of design rules. Within the negligible load envelope, the generalized loads on the component, both primary and secondary, are small enough that the component will pass any set of high temperature design rules considered here. By identifying this region, potentially for different materials, temperatures, design lives, etc., a general criterion for accepting the use of any set of design rules for a particular component can be identified.

The following analysis considers the most restrictive envelope of the four sets of design rules but ignores the fatigue criteria for the Section VIII methods. The fatigue evaluation criteria for Section VIII does not extend to the creep regime, with the exception for fatigue screening by comparable experience. A previous report [1] includes additional details and analysis of the Section VIII methods. However, the authors expect the Section III, Division 5 creep-fatigue criteria to control over a Section VIII fatigue analysis, as Section VIII does not include creep-fatigue interaction, so this caveat does not affect the general applicability of the following analysis and recommendations.

To simplify the definition of this negligible load envelope, the following assumptions are made:

1. Define the envelope as a triangle in the primary load-secondary load space. This simplifies the definition of the space, as it can be defined with just two numbers: the distance along the primary load axis (P_e) and the distance along the secondary load axis (Q_e). This is a conservative simplification (see Figure 3.2), as it conservatively misses some of the overlapping design space, particularly at lower temperatures.

2. Eliminate the cycle hold time as a parameter by selecting the most conservative points P_e, Q_e for all the hold times considered in the original study.

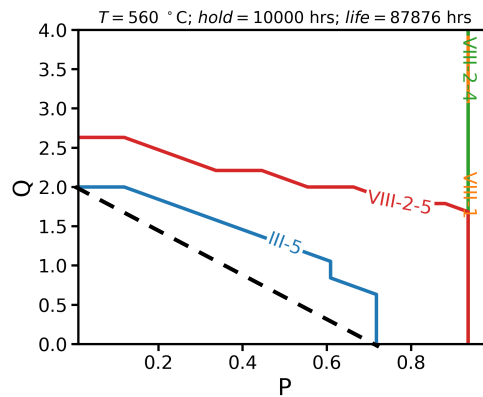


Figure 3.2. Example of drawing a conservative triangular envelope (dashed line) underneath the design-rule-specific envelopes.

After making these assumptions, a series of temperature, design life, and material specific envelopes are defined. Figure 3.3, Figure 3.4, and Figure 3.5 plot these material-temperature-life specific envelopes for 316H, Grade 22, and 304H. For these three materials, a complete comparison between the Section III, Division 5 and Section VIII rules can be made, as described in a previous report [1].

Each figure shows all the envelopes at a given temperature, for different design lives. At lower temperatures, the design life does not affect the envelope, as creep is effectively negligible. At higher temperatures, increasing the design life reduces the size of the envelope, as creep and creep-fatigue limits the life of the structure. Increasing the temperature likewise decreases the size of the envelope. Grade 22 is weaker than the stainless steels in the sense that the design life effects manifest at lower temperatures. Likewise, these plots show 304H is weaker than 316H for the same reasons: creep effects appear at lower temperatures in 304H, and design life has more of an effect on the envelope compared to 316H.

As the design life increases, the limits decrease in a staircase fashion, jumping from one value to the next making a ladder. This is an artifact of how the plots were generated, as the diagrams were created by sampling discrete values of the primary and secondary stress. Additional discussion on this procedure can be found in the literature [1].

In theory, a negligible load criterion could be defined from these diagrams. This criterion would vary with material, with temperature, and with design life. Basing the negligible load criterion off these detailed results would produce a more accurate, less conservative method, but defining and applying the criteria would be difficult.

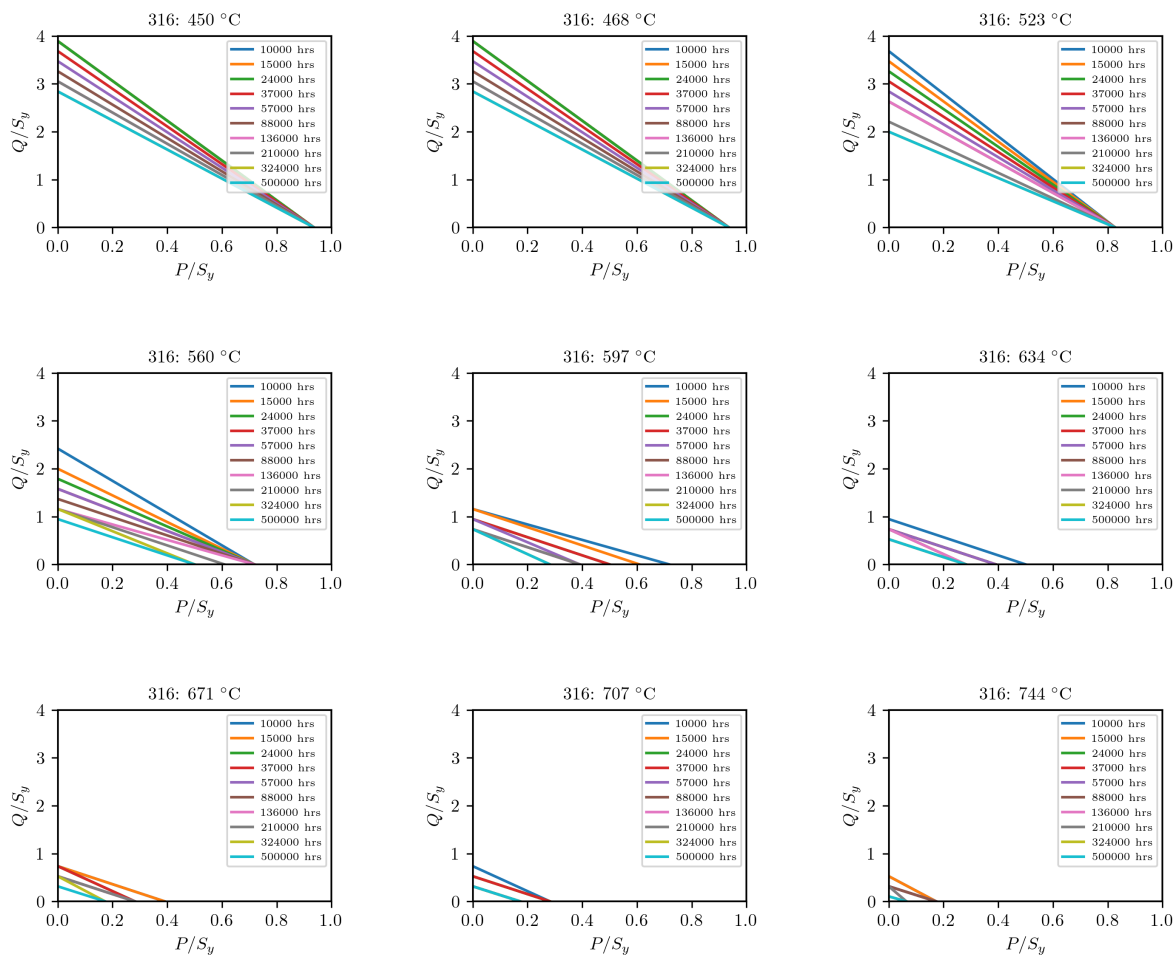


Figure 3.3. Conservative negligible load envelopes for 316H.

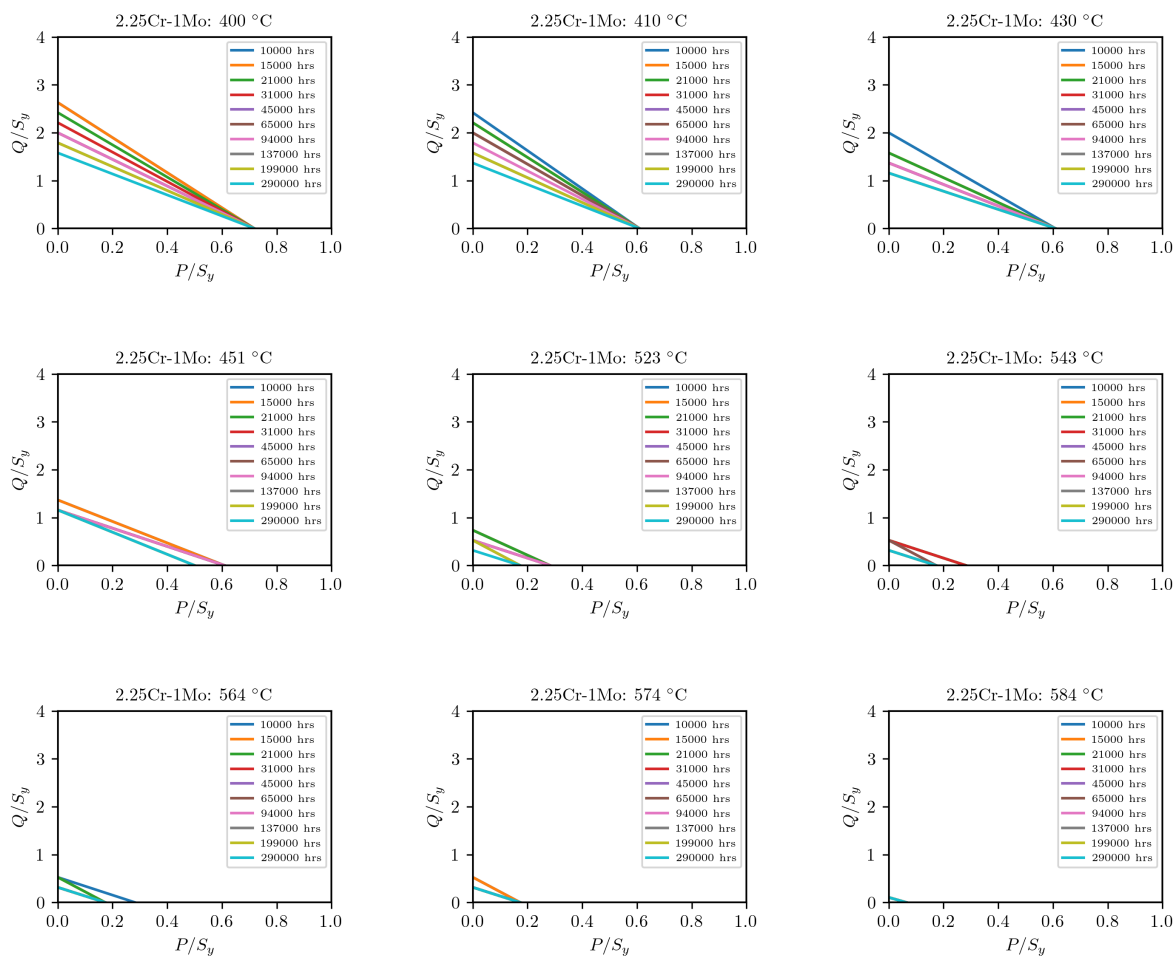


Figure 3.4. Conservative negligible load envelopes for Grade 22.

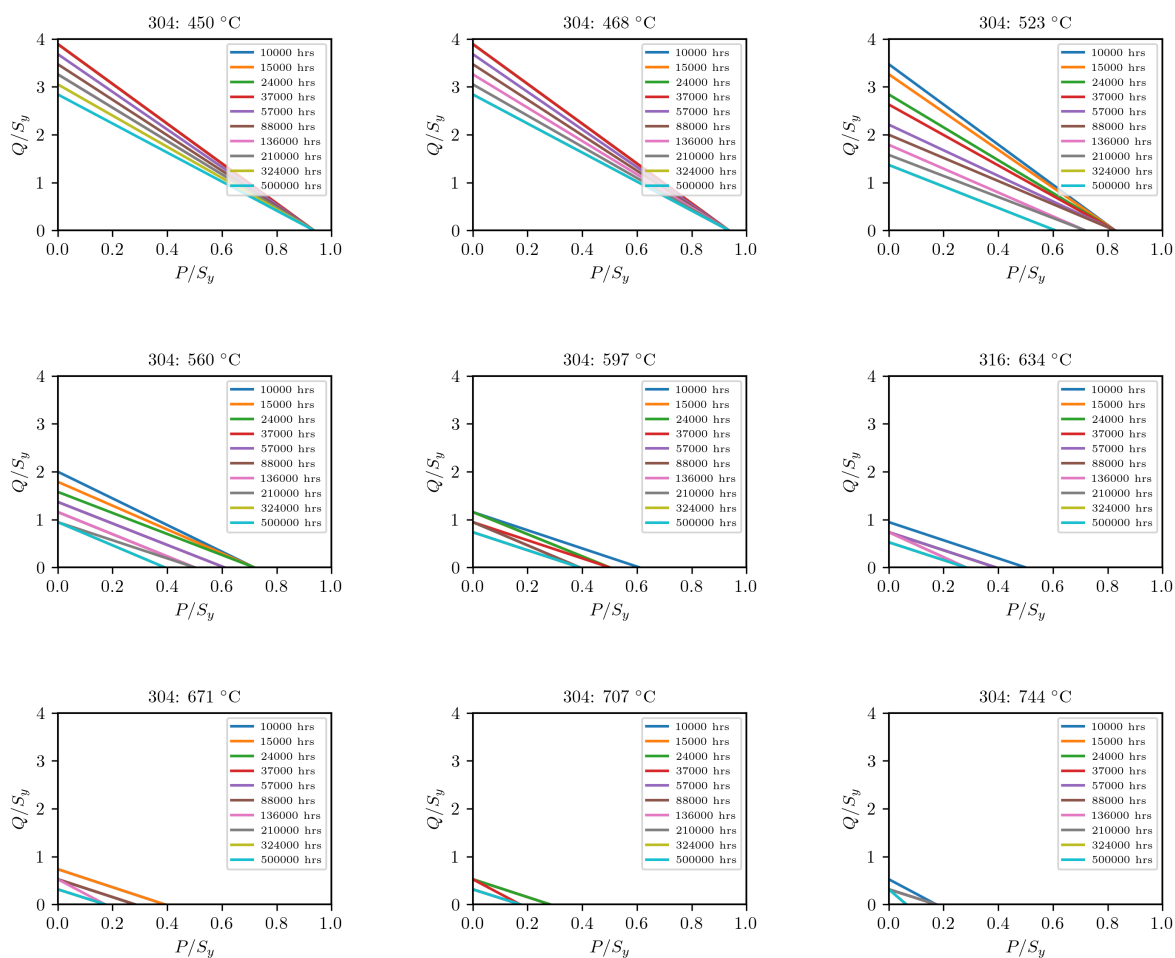


Figure 3.5. Conservative negligible load envelopes for 304H.

3.2.2 Simplified approach

As an alternative to using the full material/temperature/time specific data, replotting the same information on a different scale is proposed. Figures 3.3-3.5 plot the non-dimensionalized primary and secondary stress as the ratio of the actual primary or secondary stress to the design material yield stress S_y , with the yield strength calculated at fixed temperature. Alternatively, the same data normalized to a combination of the yield and creep rupture strength of the material could be plotted. The idea with this normalization is it could account for, or at least reduce, the time and temperature effects by including time-dependent creep data.

Figure 3.6 plots the negligible load envelopes normalized by S_{yr} , defined as the minimum of the material yield stress and the minimum stress to rupture. This value is defined as

$$S_{yr} = \min\{S_y, S_r\}$$

with the Code values of the yield strength and minimum stress to rupture. As with the yield strength, for each individual case in generating the envelopes, the strength at constant temperature is used. This is a reasonable approximation for small ΔT . However, the multiple envelopes plotted in the figure cover the entire temperature range considered for each material in the parametric Bree analysis.

The comparison includes the three materials discussed in the previous section along with two more: Alloy 800H and Grade 91. For reasons discussed in a previous report [1], the Section VIII and Section III, Division 5 rules for these two materials cannot be easily compared, so the envelopes for these two materials are based entirely on a Section III, Division 5 analysis. The comparison between the methods suggests that the Section III, Division 5 rules create the bounding envelope, so the results even for these materials can be generalized.

Normalizing by S_{yr} helps collapse the material, time, and temperature dependence, but does not fully unify the envelopes. However, a conservatively drawn universal envelope on this scale for four of the five materials can be made. This universal envelope spans from $\left(\frac{P}{S_{yr}}, \frac{Q}{S_{yr}}\right) = (0, 0.5)$ to $\left(\frac{P}{S_{yr}}, \frac{Q}{S_{yr}}\right) = (0.25, 0)$. Grade 91 is the exception to this universal envelope, with some of the higher temperature results not being bounded by this triangle. Figure 3.6 also includes this universal envelope, plotted in red.

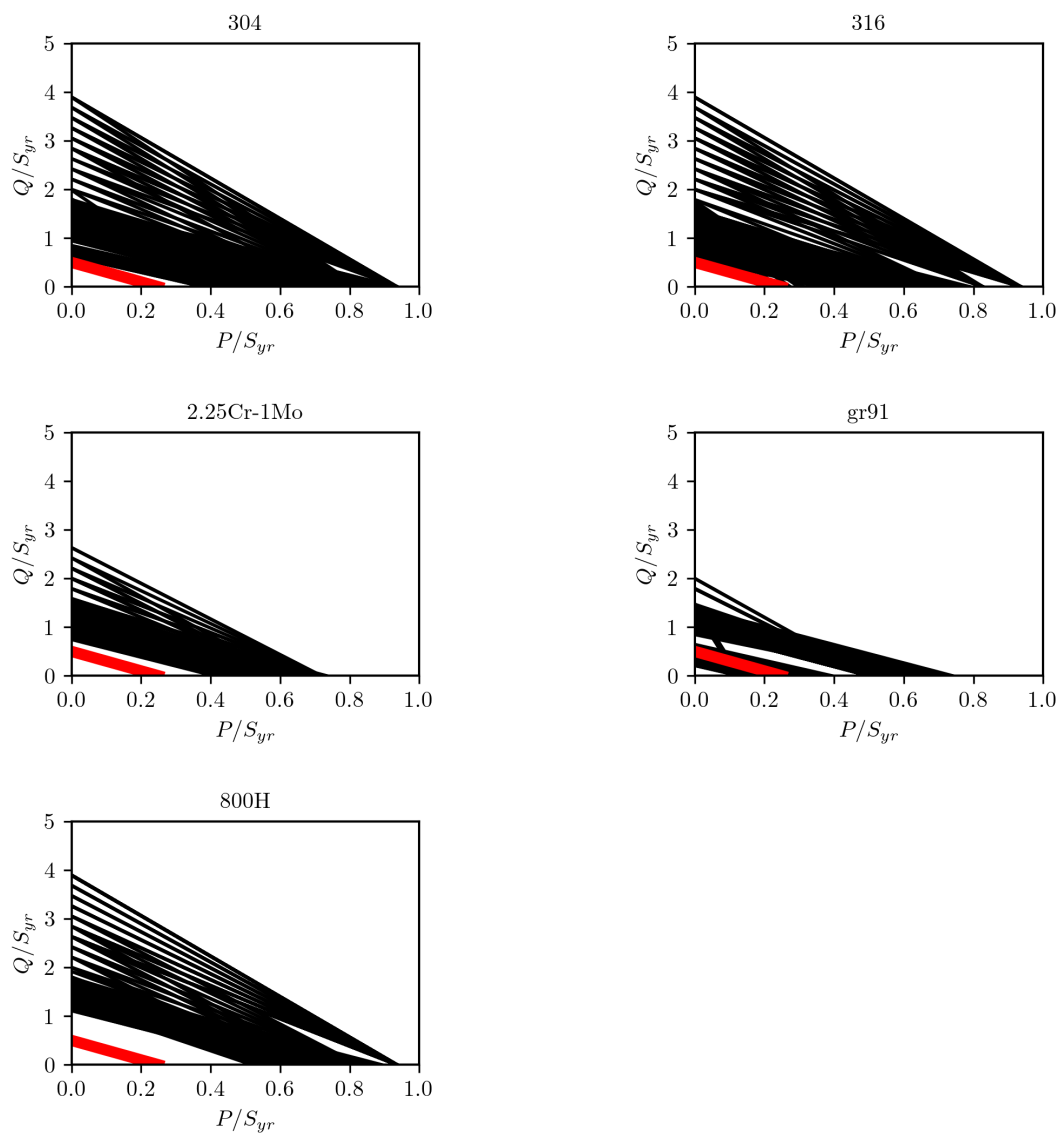


Figure 3.6. Universal negligible load envelopes normalized by S_{yr} for the five materials. The black lines are the analysis results, and the red line is the proposed universal envelope.

3.2.3 Generalizing the criteria

The results underlying this analysis were generated by analyzing many discrete Bree cylinder designs. Past work [5] [6] and the ASME Code accepts analyses of these types as general design criteria, provided peak stresses are negligible. The Code extends the Bree analysis to arbitrary structures through the process of stress classification. For each stress classification line in the component, the designer classifies the local stresses as primary, secondary, and peak and calculates the corresponding stress intensities. They then enter the non-dimensionalized values of primary and secondary stress into the Bree diagram. This approach seems reasonable for assessing negligible load, comparing the results to the negligible load bounding envelope.

The authors have no direct evidence that this approach generalizes to other materials, outside of the five considered here. Indeed, the approach does not seem to work for Grade 91, though that material tends to be the exception to many high temperature design approaches. The normalization stress S_{yr} was selected deliberately to be something that could be calculated for materials in Section VIII and not in Section III, Division 5, i.e., materials with only an ASME Section II, Part D dataset available. Without additional design studies and test data, this criterion leads to very small values of stress being allowed for all materials. At low temperatures this criterion results in total stresses that are less than a quarter of the ASME yield strength S_y , well below the typical primary stress allowable criteria of $0.67S_y$. Similarly, at high temperatures the total stress will be less than a quarter of the minimum rupture strength S_r , which again is well below the typical factor placed on the rupture data for any of the design methods considered here.

Overall, there is no direct analytical support for applying this approach to materials outside 316H, 304H, 800H, and Grade 22. However, it is highly likely the criterion will transfer to additional materials, excluding the category of creep-strength enhanced ferritic-martensitic steels, like Grade 91.

3.3 Recommendation

For 316H, 304H, 800H, and Grade 22 base metal the authors recommend a negligible load criterion of

$$4P + 2Q \leq S_{yr}$$

for locations with negligible peak stress. This formula is a restatement of the triangular design envelope described above. If a design passes this criterion, then this analysis suggests it will not matter if it is analyzed with the Section VIII or Section III, Division 5 design rules – it would pass both.

This universal envelope encompasses all the variability in the original parametric study: design life, temperature, hold time, etc. A somewhat less restrictive alternative would be to use the material and/or temperature specific envelopes shown in Figures 3.3, 3.4, and 3.5. The challenge here would be to easily parametrize and describe how the envelopes change as the temperature is varied.

4 Negligible Creep

For pure fatigue loading, without creep-fatigue interaction, the Section VIII, Division 2 fatigue design rules are fundamentally adequate for high temperature applications. However, as described in [1], which summarizes differences in the Section VIII, Division 2 and Section III, Division 5 methods for high temperature design, high temperature fatigue data are not currently available in Section VIII, Division 2. Chapter 6 discusses potential remedies for the lack of fatigue data. Assuming the availability of high temperature fatigue data tested in accordance with ASTM E606 standard, the Section VIII, Division 2 approach to fatigue design might be applied to high temperature components experiencing negligible creep-fatigue interaction. These data need not capture hold time or rate effects and could be collected at standard fast strain rates, resulting in short test durations.

Likewise, the Section VIII, Division 2 rules guarding against plastic collapse and ratcheting/excess strain accumulation reasonably screen against these failure modes at high temperatures. One route to applying the Section VIII, Division 2 rules without restriction is therefore to screen components for creep-fatigue interaction. If creep-fatigue interaction is negligible the designer could proceed by following the current Section VIII rules.

The simplest approach to screening out components experiencing creep-fatigue interaction is to eliminate either significant creep or significant fatigue. Components inside these limits, illustrated in Figure 4.1 by reference to a Section III, Division 5 creep-fatigue interaction diagram, would clearly not experience creep-fatigue interaction. This chapter explores criteria for negligible creep; the next chapter considers negligible fatigue.

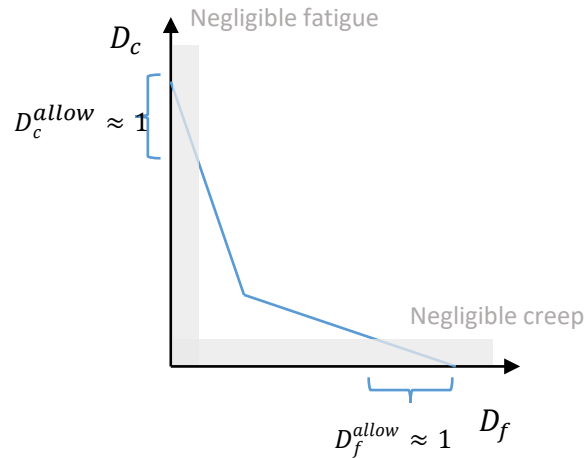


Figure 4.1. Schematic illustrating criteria for negligible creep and fatigue on a Section III, Division 5 style D-diagram.

4.1 Negligible creep criteria

Negligible creep implies the component will operate in the creep regime and will experience creep effects, i.e. strain accumulation and/or stress relaxation, but these creep effects will not significantly influence its cyclic life. Negligible creep does not exclude service in the creep regime, only limiting creep effects to be so small enough as to be safely neglected.

A definite criterion could be established from the Section III, Division 5 creep-fatigue interaction diagrams, as illustrated in Figure 4.1. The approach is:

1. Select an allowable fatigue damage fraction, D_f^{allow} . This is a free choice, selecting a higher value will produce a more stringent creep limit but make the fatigue evaluation easier to meet. The only change to the fatigue design method will be to restrict the allowable fatigue damage to D_f^{allow} , rather than 1.0.
2. Index D_f^{allow} in the material-specific D-diagram to find a corresponding value of $D_c^{negligible}$.
3. Utilize the negligible creep screening method to ensure the creep damage fraction in the component remains less than $D_c^{negligible}$.

This approach is rigorous and is backed by the creep-fatigue tests underlying the Section III, Division 5 D-diagrams. It is also limited to the materials with available D-diagrams in the Code, i.e. the Section III, Division 5 Class A materials.

Table 4.1 calculates the values of $D_c^{negligible}$ for several choices of D_f^{allow} .

Table 4.1. Negligible creep damage fractions for various choices of material and allowable fatigue.

	D_f^{allow}		
	0.7	0.8	0.9
316H	0.13	0.09	0.04
304H	0.13	0.09	0.04
Grade 91	<0.01	<0.01	<0.01
Grade 22	0.03	0.02	0.01
800H	0.03	0.02	0.01
Alloy 617	0.03	0.02	0.01

Except for Grade 91, while the results vary by material and by the selection of the allowable fatigue damage, values for the negligible creep criteria range from 0.01 to 0.13. These values seem to align well with existing approaches. For example, the Section III, Division 5, Subsection HB, Subpart B HBB-T rules for negligible creep, discussed in greater detail below, uses a creep damage criterion of 0.1. The negligible creep damage fraction for Grade 91 in Table 4.1 is low relative to the other class A materials because of the stronger creep-fatigue interaction in the material.

4.2 Existing approaches

4.2.1 Section VIII

Section VIII does not directly reference the concept of negligible creep. However, the Section VIII dividing line for creep effects is typically when the allowable stresses in Section II, Part D are controlled by the time-dependent properties. The stress tables indicate this condition with italicized values.

This condition means Section VIII effectively uses a *negligible* creep, rather than a strict *no* creep criteria for their design rules. The Section VIII allowable stresses are based on extrapolated 100,000-hour properties. The point at which the 100,000-hour creep properties control in the allowable stress over the tensile properties is well into the creep regime for all materials. However, this approach does not invoke the actual stresses in the component. As such, it may be possible to further extend the Section VIII rules into the creep regime, even with negligible creep, by considering the stress and creep damage fraction in the criteria.

4.2.2 Section III, Division 5

Section III, Division 5, Subsection HB, Subpart B includes negligible creep rules as the A-3 test in HBB-T-1324. There are three criteria, all of which must be satisfied for the component to be deemed to operate in the negligible creep regime:

1. HBB-T-1324(a): A time-fraction creep criterion with an assumed stress and (potentially) a safety factor on the creep damage. Except for Grade 91, the assumed stress is 1.5 times the design yield stress (which is somewhat less than 1.5 times the actual material yield strength) with no safety factor on the damage. For Grade 91, the assumed stress is the design yield strength, which accounts for the lack of work and cyclic hardening for the material, and a substantial safety factor of 10 on the creep damage. The creep damage fraction deemed to be negligible is 0.1.
2. HBB-T-1324(b): A criterion related to creep strain accumulation at an assumed stress of 1.25 times the design yield strength. The strain accumulation damage is 0.2%, likely by analogy to the standard offset strain used in determining yield strength from tensile test results. The rules do not specifically call out a source of creep deformation data, but given the context, the material isochronous stress-strain curves would likely be used.
3. HBB-T-1324(c): A shakedown criterion based on a modification of the Section III, Division 1 approach. The modification accounts for the reduction in the shakedown limit caused by stress relaxation on the hot end of the cycle. This criterion relates more to the purpose of the A-3 check as a method for assessing the component against the HBB-T strain accumulation and ratcheting limits, rather than negligible creep directly.

The first two checks are reasonable ingredients for a negligible creep criterion, covering both creep damage and strain accumulation. However, these criteria are very conservative. The assumed stresses for the creep damage and strain accumulation are high, as typically high temperature designs will operate well below even the design material yield stress to avoid excessive creep-fatigue damage and ratcheting strain accumulation. Calculating the criteria with an assumed stress is convenient as it avoids requiring even a preliminary stress analysis of the component but will restrict negligible creep components to those operating only for very short times at moderately high temperatures.

4.2.3 Recommendations

A component that passes the HBB-T-1324 A-3 (a) and (b) criteria could be deemed to have negligible creep-fatigue damage and therefore be designed to the Section VIII rules without additional modifications, provided high temperature fatigue data is available. For materials other than creep-strength enhanced ferritic (CSEF) steels, the $r = 1.0$ and $s = 1.5$ factors applied in Section III, Division 5 for the non-Grade 91 materials could reasonably be extended to other non-Class A materials. For CSEF materials, the Grade 91 values might be used instead. As noted above, these are conservative factors even in the context of the Section III, Division 5 rules.

The creep rupture and deformation information required to execute the two criteria are not readily available for non-Section III, Division 5, Class A materials. One compromise might be to evaluate the creep strain accumulation check with the minimum creep rate, rather than isochronous curves. This reduces the conservativeness of the check, as the primary creep rates accounted for in the isochronous curves are faster than the minimum creep rate. However, this approach can be justified by the extreme conservativeness in the assumed stresses and strain limits. In theory, creep rupture and minimum creep rate data are available for all Section VIII materials as they factor into the allowable stress definition.

Additional work could reduce the conservatism of the general criteria by including some measure of the actual stress in the component in the negligible creep deformation. Chapter 3 of this report on negligible load might offer a path forward along these lines. Ideally, new criteria might require only rupture and minimum creep rate data, as this would make it easier to apply the new method to materials outside of the Section III, Division 5, Class A materials.

5 Negligible Fatigue

Negligible fatigue is the converse of negligible creep: a component where there is no significant creep-fatigue interaction because the fatigue damage fraction is so small (Figure 4.1). An analogous argument can be drawn for this condition as for negligible creep; as there is insignificant creep-fatigue interaction, the Section VIII design rules, which do not account for creep-fatigue, may be adequate without additional checks.

5.1 Negligible fatigue criteria

A detailed criterion for negligible fatigue can be determined analogously to the negligible creep criterion described in Section 4.1. The process is simply reversed from the procedure in that section; choose an allowable creep fraction and then use the Section III, Division 5 creep-fatigue interaction diagrams to find the corresponding allowable fatigue damage. Table 5.1 summarizes the results for the six Class A materials. Except for Grade 91, the results are nearly identical to the negligible fatigue results because the D-diagrams are symmetric. Again, a universal value of $D_f^{negligible}$ between 0.01 and 0.13 is reasonable for all of the materials except Grade 91.

Table 5.1. Negligible fatigue criteria for different choices of allowable creep damage.

	D_c^{allow}		
	0.7	0.8	0.9
316H	0.13	0.09	0.04
304H	0.13	0.09	0.04
Grade 91	0.03	0.02	0.01
Grade 22	0.03	0.02	0.01
800H	0.03	0.02	0.01
Alloy 617	0.03	0.02	0.01

5.2 Existing approaches

5.2.1 Section VIII

Section VIII, Division 2, Part 5 includes several negligible fatigue checks. These are contained in Section VIII, Division 2, Section 5.5.2. These include two methods (Method A, 5.5.2.3 and Method B, 5.5.2.4) based on pressure and temperature cycle counting, the component geometry, and (low temperature) fatigue data (for Method B). Method A is limited to materials with tensile strengths below 552 MPa; Method B is general.

However, neither Method A nor Method B are allowed for components operating in the creep regime. In this case, only the comparable experience criteria in 5.5.2.1 applies. This provision exempts components from fatigue analysis when similar equipment exists and has operated successfully in a similar cyclic environment. The provision includes a list of features to consider when evaluating similarity but is otherwise a qualitative, rather than quantitative, approach.

In the near term, this qualitative exemption will be difficult to apply for components intended for high temperature reactors as comparable equipment generally does not exist. A more quantitative approach would be useful for high temperature reactor components.

5.2.2 Section III, Division 5

There are currently no negligible fatigue design checks in the Section III, Division 5 rules.

5.3 A new method

This section describes a qualitative approach for classifying a component as having negligible fatigue. The method is general, i.e. it could be applied to any material and component, but relies on material data that is not generally available outside of the Section III, Division 5, Class A materials, specifically:

1. High temperature continuous cycling fatigue curves.
2. A creep-fatigue interaction diagram.

The second data requirement could be removed by assuming a universal interaction diagram; for example, a diagram with an intersection point of (0.2,0.2). However, additional test data may be needed to verify the choice of the universal intersection point and any universal D-diagram would need exceptions (for example, for creep-strength enhanced ferritic-martensitic steels). The first requirement is necessary, as different materials will have different fatigue strength and consequently different negligible fatigue limits. However, some general conclusions could be drawn by conservatively examining the limits produced for the six Class A materials, treating them as a representative sample of broader material categories.

The method has two steps:

1. Demonstrate the component will shake down elastically. Any method could be used to demonstrate elastic shakedown, for example, the B tests in Section III, Division 5, Subsection HB, Subpart B, Nonmandatory Appendix HBB-T or the methods in Section VIII, Division 2, Section 5.5.6. The Section III, Division 5, B tests could not be used as written but rather modified to require the component to operate in the elastic shakedown regime.
2. Determine a negligible cycle count, which might be material specific. This criterion conservatively assumes all alternating load events cause the maximum possible strain range allowed by the elastic shakedown requirement.

The remainder of this section focuses on defining the second criterion.

Ignoring the effects of creep, under elastic shakedown the strain range experienced by the component at any location in the steady cyclic limit, i.e. after achieving elastic shakedown, is limited by the maximum possible strain range across the von Mises yield surface. Adopting the equivalent von Mises strain as the applicable strain measure and simplifying, this strain range is given by

$$\Delta\epsilon_{max} = \frac{2\sigma_y}{3\mu}$$

with σ_y the material yield stress and μ the shear modulus. This can approximate the actual material yield stress with the ASME design yield strength S_y and calculate the shear modulus by reference to the Young's modulus E and Poisson's ratio ν values given in Section II, Part D of the ASME Code. This gives the expression

$$\Delta\epsilon_{max} = \frac{4S_y(1 + \nu)}{3E}$$

Creep strain caused by steady holds during the cycle could increase this strain range. This additional straining could be estimated with the Section III, Division 5 isochronous curves or with minimum creep rate data, assuming the effective creep stress is equal to the yield stress. However, except at very high temperatures, the creep strains produced by this correction will be moderate because the stress is limited by the yield surface and the cycle hold time is typically a fraction of the overall design life. The subsequent analysis does not make a creep correction to the strain range, though further work might examine the effects of creep strain in greater detail.

The strain range can be calculated for any material. To determine the allowable cycles, first consider the analysis in Section 5.1. For a choice of allowable creep damage, the D-diagram can be used to find a corresponding value of allowable fatigue damage, D_f . With this value, the value of the maximum strain range, and a strain-based fatigue diagram for the material, the maximum allowable number of cycles as a function of temperature can be calculated.

Figure 5.1 plots the results for the six Class A materials for the material-specific D-diagrams found in Section III, Division 5, using the factored, design fatigue curves from Section III, Division 5, HBB-T, using an allowable creep damage fraction of 0.8.

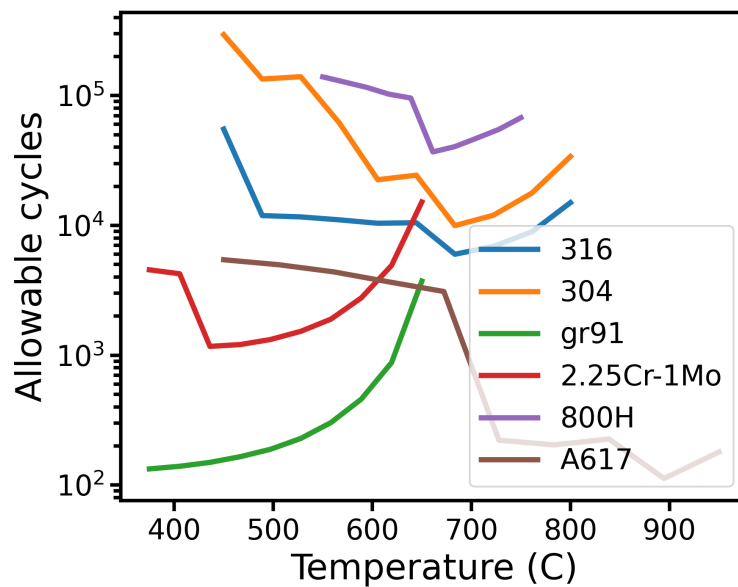


Figure 5.1. Negligible fatigue criteria for the six Class A materials as a function of temperature.

The results vary greatly both in terms of material and temperature dependence. The austenitic steels and 800H follow a general trend of decreasing allowable cycles as temperature increases for lower temperatures before transitioning to increasing allowable cycles as temperature increases at higher temperatures. However, except for 304H, the variation in the allowable cycles over the temperature range considered here is small for these three materials. The allowable cycles for ferritic steels essentially monotonically increase as temperature increases, not counting a small region of temperatures for Grade 22. The allowable cycles for the nickel-based Alloy 617 decrease as temperature increases.

The temperature dependence of the maximum shakedown strain range is a function of the temperature dependence of the material yield stress and elastic properties. This strain range is approximately constant for the austenitic steels and 800H, decreases with temperature for Grade 91 and Grade 22, and increases with temperature for Alloy 617. This temperature dependence of the strain range mostly explains the general temperature dependence in the allowable cycles noted above.

To avoid temperature-dependent criteria, the minimum value for each material is selected. Table 5.2 presents these results, rounded conservatively to the nearest half order of magnitude.

Table 5.2. Negligible fatigue cycles.

Material	316H	304H	Grade 91	Grade 22	800H	Alloy 617
N_f	5,000	10,000	100	1,000	10,000	100

It is somewhat difficult to generalize from this table. Grade 91 is oftentimes an exception to general guidelines in the Code, owing to its metastable structure. However, Alloy 617 also exhibits a low negligible cycle threshold. For Alloy 617 the drop in the criterion occurs around 700° to 800°C, where again the material undergoes a microstructural change notable in other experimental data. With these two exceptions, a general rule of 1,000 cycles seems adequate, perhaps extending to 5,000 cycles for austenitic steels. For Grade 91 and Alloy 617 at high temperatures, a limit of 100 cycles may be required.

Quantitatively extending this criterion to other materials, e.g. those qualified for Section VIII but not for Section III, Division 5, requires both high temperature fatigue and creep-fatigue data. The qualitative values given in the previous paragraph could be more broadly applied to materials in similar categories (for example other stainless steels for 5,000 cycles, other low alloy ferritic steels for 1,000 cycles).

5.4 Recommendations

More work on negligible fatigue criteria would provide greater confidence in the values provided by the analysis in Section 5.3. Ideally, these methods would be codified in Section III and/or Section VIII. One barrier to more quantitative criteria is the lack of high temperature continuous cycling fatigue data, which would also benefit the Section VIII fatigue approach in general.

6 Modifications to Section VIII to Address Creep Strength of Weldments

6.1 Background

The general ASME philosophy for weldments in low temperature design is to ensure they are overmatched compared to the base material. That is, the tensile strength and, if relevant to the design, fracture toughness of the weldment exceeds the strength of the corresponding base metal. ASME design methods for low temperature construction apply this concept to essentially ignore the welds in the component sizing for design by rule or component analysis for design by analysis. This is not to say the design and construction rules ignore weldments entirely: there are detailed requirements for weld geometries, inspections, types, etc.; simply that at low temperatures, the design methods assume the weldment will be stronger than the base metal and neglect the weldments in the detailed analysis and sizing calculations.

Welds that are overmatched for low temperature, time-independent properties are nearly always undermatched for high-temperature properties, including creep and creep-fatigue strength. This fact has been demonstrated for a wide variety of materials including:

1. CSEF materials, including Grade 91 (e.g. [7, 8, 9, 10, 11, 12, 13] among many others). The reduced strength of weldments, attributed to Type IV cracking, has severely impacted the service performance of this category of materials.
2. Austenitic stainless steels, including types 316, 304, 308, and 347 [14, 15, 16, 17, 18].
3. Alloy 800H [19, 20].
4. Nickel-based alloys, including Alloys 617, 740H, and 282 [21, 22, 23, 24].

These encompass most materials employed in elevated-temperature construction – the reduced creep and creep-fatigue strength of weldments seems to be a universal phenomenon.

At least two factors contribute to the lower measured strength of a weldment compared to base material [16]:

1. The metallurgical notch effect. The weldment often has different inelastic deformation properties, i.e. yield strength, work hardening, cyclic hardening, and creep rate, compared to the base material. This difference in properties can cause a stress concentration.
2. Reduced resistance to cavitation and other creep and creep-fatigue damage mechanisms.

While the inelastic properties of weldments often differ from base material, the elastic properties are often similar. This means the metallurgical notch effect only occurs at high temperatures, where service loads cause the material to yield and creep, and in low cycle fatigue. In low temperature service or in high cycle fatigue conditions where the material behavior is essentially elastic, the metallurgical notch effect is insignificant and weldments often have similar measured strengths to base material.

The measured strength of a cross-weld creep sample, encompassing both weld-material and the heat affected zone (HAZ) of the base material, is often different from samples built up from just weld metal [16, 25]. Modern work establishing the creep strength and allowable stresses of welds focuses on cross-weld samples, though earlier work includes test data on built-up weld-material samples.

6.2 Sources for weld strength reduction/stress rupture factors

Section VIII does not adjust the strength of welds to account for the reduced creep and creep-fatigue strength of weldments at elevated temperatures. Instead, Section VIII considers a joint efficiency factor E . This factor is not intended to account for the reduced strength of weldments at elevated temperatures as it: (1) does not vary with temperature; (2) is not indexed to weldment creep test data. Rather, the joint efficiency factor aims to account for defects in weldments that may not be identified through inspection. The factor therefore varies with weld type and inspection technique, but not temperature.

However, both Section III, Division 5, and Section I of the ASME Boiler Code do make such adjustments.

6.2.1 Section III, Division 5

Section III, Division 5 applies the adjustment via stress rupture factors, R . These factors adjust the Code minimum stress to rupture for the base material, S_r . The factors can depend on both temperature and time of service and are specific to specific combinations of base material, weld material, and welding technique.

In the past, the Code did not document the criteria for calculating the stress rupture factors, though Nonmandatory Appendix Y did specify they be based on cross-weldment specimens including weld metal, the HAZ, and base metal (and further requires detailed specification of the weld type). The 2025 edition of the Code will specify that the factors be calculated as the ratio of the average weldment creep rupture strength to the average base metal creep rupture strength, which was the undocumented but accepted approach used to calculate the current factors. The factors are capped at $R = 1$, though typically this ratio will always be less than 1 at high temperatures. Conventionally, these values are calculated with Larson-Miller fits to the cross-weldment and base metal creep rupture data.

Applying the factors in the design calculations is straightforward as they index to the base metal minimum stress to rupture. Calculations using S_r instead use the product $R \times S_r$. The Section III, Division 5 allowable stresses reference the minimum stress to rupture so they can be applied to weldments by considering the minimum of the base metal allowable stress and the product of the appropriate safety factor times $R \times S_r$. Note that Division 5 adjusts the safety factors in the definition of the allowable stresses, for example using 80% of $R \times S_r$ for Service Level A and B loadings rather than 67% S_r for base metal.

6.2.2 Section I

Section I provides Weld Strength Reduction Factors (WSRF) applied to base-metal allowable stresses in calculating the maximum pressure or minimum thickness of a component. For welded joints these factors *replace* the joint efficiency factor (PG-27.4.1), though Section I retains the concept of joint efficiency for other types of connections, for example riveted joints. Table PG-26 provides these factors which are temperature dependent, but not time dependent, and cover a wide variety of materials. These factors are based on 100,000-hour creep properties, which is the same as the underlying base-metal allowable stresses. The factors apply to cylindrical components with longitudinal seam welds as well as spherical sections with welded segments (including spherical heads) and only applies to steady loads. A recent ASME report provides some details on how the factors are calculated [26].

Section I simply applies the WSRF, w , directly to the allowable stress. That is, a welded joint uses an allowable stress of $w \times S$. This, conservatively, applies the factor even if the allowable stress might be controlled by time-independent properties at a given temperature.

6.3 Modifications to the Section VIII design procedures

6.3.1 How to apply a factor

Once the requisite factors have been determined, implementing a WSRF in the Section VIII procedures would be comparably straightforward. For the design by rule approaches, the current approach is to apply the weld joint efficiency factor, E , to the allowable stress. For the modified approach for elevated-temperature service, the WSRF would likely replace E or, conservatively, adopt the approach of applying the lesser of the weld strength reduction factor and the appropriate existing joint efficiency factor. The rationale here is that the tested welds to establish the WSRF would ideally have the same quality and level of inspection as the corresponding service welds, so the tests would directly characterize the weldment strength.

The Section VIII, Division 2, Part 5 design by analysis rules do not currently incorporate the efficiency factor. The WSRF would be added by multiplying the allowable stress, S , in the design by analysis criteria with the appropriate reduction factor.

There are two options for when to apply the factors:

1. Follow Section I practice and simply always apply the factor regardless of the design temperature.
2. Only apply the factor in the creep regime. A simple criterion would be to apply the factor only when the allowable stress tables in Section II, Part D indicate the allowable stress is based on time-dependent properties (i.e. when the values are italicized).

The second approach seems to reasonably balance the need for a correction (i.e. only in the creep regime) and the difficulty of implementing a factor in the Section VIII rules.

As noted in Section 6.1, the current Section VIII, Division 2, Part 5 design by analysis rules include specific modifications to account for the fatigue strength of welded connections. While this report does not specifically validate this methodology, the current approach seems reasonable. However, as noted in previous chapters of this report, the method currently does not encompass the fatigue strength of welds in elevated-temperature service. Applying the method to elevated-temperature construction would require supplementing the Section VIII fatigue data with information on the high-temperature strength of weldments. Potentially this data could be taken, for a limited number of materials, from Section III, Division 5.

6.3.2 *Where to source weld factors*

Without needing to develop a new methodology, a modified Section VIII approach could source weld factors from either Section I or Section III, Division 5. For Section I, the factors in Table PG-26 could be used. For Section III, Division 5, the stress rupture factors at 100,000 hours could be taken to be the corresponding Section VIII correction. Table PG-26 covers additional materials, compared to Section III, Division 5, and already normalizes the factors to 100,000 hours of service and so this source may be the better option.

6.4 Differences in weld inspection requirements

As noted in Section 6.2, the Section VIII weld joint efficiency factor varies based on the weld type, the material type (for Section VIII, Division 2), and the type and degree of inspection. As such, comparing the current Section VIII joint efficiency factors to the Section I WSRF and the Section III, Division 5 stress rupture factors also require examining the type and degree of inspection required by each section of the Code.

All three sections of the Code require weld qualification per Section IX of the Code. Inspection requirements discussed here are for the actual welded component.

The Section III, Division 5 weld examination requirements for Class A components are in HBB-5200. The general approach is a double volumetric inspection using radiography plus one of ultrasonic inspection, eddy current examination, or a second radiographic examination at a different angle. Smaller and less critical welds have reduced inspection requirements. Typically, this reduced requirement is a single radiographic inspection.

PW-11 provides the general weld inspection criteria for Section I, and PW-46 through PW-53 document the detailed Section I inspection requirements for welded construction. In general, welded butt joints require volumetric inspection by either ultrasound or radiography, though Table PW-11 provides exemptions for circumferential joints depending on thickness. The Code provides acceptable weld geometries for nozzles and other connections but does not require volumetric inspection.

The Section VIII requirements are the most diverse, as the Code gives the designer a range of options to choose from for each connection type. The idea is that the designer can use a lower, more conservative, joint efficiency factor to make up for reduced inspection. Table UW-12 provides the options for Section VIII, Division 1 and Table 7.2 for Section VIII, Division 2. Division 1 requires volumetric inspection by radiography or ultrasound, Division 2 provides options for both volumetric and surface inspection (with different efficiency factors).

Summarizing is difficult given the range of weldments considered by the tables. For longitudinal butt joints, Division 1 provides full efficiency ($E = 1$) for fully inspected joints, $E = 0.85$ for spot inspection, and $E = 0.70$ for no inspection. The efficiency factor for a comparable joint in Division 2 ranges from 1.0 for full inspection to 0.85 for 10% volumetric inspection.

Comparing the inspection requirements for longitudinal butt joints is perhaps the easiest way to directly compare the rules:

Section III, Division 5, Class A:	full double volumetric inspection by radiography plus an additional technique
Section I:	full single volumetric inspection by either radiography or ultrasonic inspection
Section VIII:	varies from full single volumetric inspection to no inspection, depending on the selected efficiency factor

Even for the most stringent inspection requirements, both Section I and Section VIII are less rigorous than the double volumetric requirement in Section III, Division 5. The Section VIII approach is the most flexible in that it gives designers a choice in trading off inspection for reduced design strength. However, note that Section VIII allows designing a longitudinal butt joint with the full strength of the base material if the joint undergoes a full volumetric inspection, while both Section I and Section III, Division 5 would *both* require volumetric inspection and take a strength reduction in the design.

7 Conclusions

This report describes several potential modifications or additional design checks that could be added to the ASME Section VIII rules to provide greater confidence in the long-term reliability of components in high-temperature nuclear reactors. These modifications address several considerations identified in the current Section VIII rules, summarized in Chapter 1.

The modifications and additions attempt to cover as many materials as possible, though detailed high-temperature cyclic data is only available for a few materials. The specific analyses in this report focus on these materials, which are those qualified for Class A construction via Section III, Division 5. However, the methods themselves are general and could be applied to other materials provided relevant test data is collected. Moreover, many of the specific techniques and general conclusions could be generalized to cover broader material categories encompassing high temperature materials like those for which the detailed test data is available.

The additional checks and modifications are simple to implement in the context of the current Section VIII rules. However, they only cover a limited subset of the high-temperature design space, essentially avoiding components with significant creep-fatigue interaction. In the short term, Code Case 2843 allows for stamping a Section VIII vessel designed with a simplified version of the Section III, Division 5, Class A design rules. This Code Case provides an existing option for high temperature creep-fatigue design in Section VIII, albeit only for the five existing Section III, Division 5, Class A materials in the 2023 edition of the Code.

Longer term, the authors are aware of efforts by the cognizant Code Committees to add creep-fatigue design rules to Section VIII (and Section I) of the ASME Boiler & Pressure Vessel Code. Currently these efforts are focusing on comparative analysis between the new approaches for Sections I and VIII using the well-characterized Section III, Division 5, Class A materials. Future work will focus on developing methods that could be extended to cover other materials where extensive creep-fatigue and fatigue data does not exist. The Section III, Division 5 Code Committees are also in the process of developing new HCB design rules for Class B components using a “universal” D-diagram to avoid the need for dedicated creep-fatigue testing and thereby expanding the range of available materials. These approaches address many of the limitations identified in Chapter 1 and obviate the supplemental design checks proposed here.

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